

 Did You Know?

Knowledge Representation (KR) helps AI systems store information in a structured way so they can reason and make intelligent decisions.



What is Knowledge Representation?

Knowledge Representation (KR) is the process of organizing and storing knowledge so that an AI system can understand, reason and solve problems effectively.



Why do we use Knowledge Representation?

- Store information efficiently
- Support reasoning
- Solve problems intelligently
- Improve decision making
- Make AI systems smarter



Types of Knowledge Representation

Method	Description
Logic	Represents facts using logical statements.
Production Rules	Uses IF–THEN rules.

Semantic Network	Represents knowledge using connected nodes.
Frames	Stores knowledge in structured records.

Logic

Logic represents facts and relationships using true or false statements. It helps AI systems perform reasoning.

Production Rules

Production Rules use IF–THEN statements to make decisions.

```
IF Temperature > 40°C
```

```
THEN Turn ON Fan
```



Semantic Networks

Semantic Networks represent knowledge using nodes and links. Each node represents an object, while links represent relationships.

Animal

↓

Bird

↓

Sparrow



Frames

Frames organize knowledge into records containing attributes and values.

Student

Name : Noor

Age : 20

Advantages

- Easy knowledge storage
- Supports intelligent reasoning
- Better problem solving
- Improves AI performance
- Organized information

Common Mistakes

- Confusing Frames with Semantic Networks
- Writing incorrect IF–THEN rules
- Ignoring logical relationships

- Using incomplete knowledge



Interview Questions

Q1: What is Knowledge Representation?

It is the method of storing knowledge so AI systems can understand and reason.

Q2: What are Production Rules?

Rules based on IF–THEN conditions.

Q3: What is a Semantic Network?

A graphical representation of knowledge using nodes and links.

Q4: What are Frames?

Frames are structured records used to store information.